Tutorial : 1

1) A small farming system protects wheather a plant will grow well (1) or poorly (0) based on daily sunlight hours and water amount in liters.

Sunlight Hours (x_1)	Water liters (x_2)	Bias (x_3)	Target [good growth = 1, poor = 0]
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

1) mean of x_1

$$u_{x_1} = \frac{3+8+5+10+7}{5}$$

$$= \frac{33}{5} = 6.6$$

2) Mean of x_2

$$u_{x_2} = \frac{3+4+2+5+3}{5}$$

$$u_{x_2} = 3$$

→ variance $= \sigma^2 = \frac{1}{N} \Sigma_i (x_i - \mu)^2$

N → Total data Inputs

x_i → Input Value

μ → Mean/average

em
unt
poorly (0)
t
nt in

Target
[good growth
r = 0] =1
0

1

0

1

1

ean of
× 2
3 + 4 + 2
+ 5 + 3
5

3

2
→ Mean/
average.

$$z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

$$z \begin{cases} \geq 0 \; ; 1 \\ < 0 \; ; 0 \end{cases}$$

$$\boxed{z_1 = w_1 x_1 + w_2 x_2 + w_3 x_3}$$

$$w_1 = w_2 = w_3 = 0 \; , \eta = 0.1$$

$\boxed{x_0 w_0 (3, 1, 1)}$

$$z_1 = (0)(3) + (0)(1) + (0)(1)$$
$$= 0$$

$$z_1 = (\hat{y}) = = 0$$

$$\boxed{\hat{y}_1 = 1}$$

Here, $\hat{y}_1 = 1$ and $y = 0$

$$\boxed{\hat{y}_1 \neq y}$$

So, update weights.

$$w_i^{new} = w_i^{old} + \eta (y - \hat{y}) x_i$$

$$w_1^{new} = 0 + 0.1 (0 - 1)(3)$$
$$= 0.1(-3) = -0.3$$

$$w_2^{new} = 0 + 0.1 (0 - 1)(1)$$
$$= -0.1$$

$$w_3^{new} = 0 + 0.1 (0 - 1)(1) = -0.1$$

row (8, 4, 1) $w_1 = -0.3$ $y = 0$

$\eta = 0.1$ $w_2 = -0.1$

$w_3 = -0.1$

$z_2 = x_1 w_1 + x_2 w_2 + x_3 w_3$

$= (-0.3)(8) + (-0.1)(4) + (-0.1)(1)$

$= -2.9$

$\hat{y} = z_2 < 0$

$\boxed{\hat{y} = 0}$

Here, $\boxed{\hat{y} \neq y}$ update weights.

$w_1^{new} = w_1^{old} + \eta(y - \hat{y})x_1$

$= -0.3 + 0.1\,(0 - 0)(8)$

$= 0.5$

$w_2^{new} = -0.1 + 0.1\,(1 - 0)(4)$

$= 0.3$

$w_3^{new} = -0.1 + 0.1\,(1 - 0)(1)$

$= 0$

row w_3 (5, 2, 1) ; η

$w_1 = 0.5$

$z_3 = w_1 x_1 +$

$= 0.5(5)$

$= 3.1$

$\hat{y} = z_3 > 0$

$y = 0$; $\hat{y} =$

$\boxed{y \neq}$

$w_1^{new} =$

$=$

$=$

w_2^{new}

w_3^{new}

$=$

$\Rightarrow w_3(5,2,1)$; $\eta = 0.1$

$w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0$, $y = 0$

$z_3 = w_1 x_1 + w_2 x_2 + w_3 x_3$

$= 0.5(5) + 0.3(2) + 0(1)$

$= 3.1$

$\hat{y} = z_3 > 0$; $\boxed{\hat{y} = 1}$

$y = 0$; $\hat{y} = 1$

$\boxed{y \neq \hat{y}}$ update weights.

$w_1^{new} = w_1^{old} + \eta(y - \hat{y}) x_1$

$= 0.5 + 0.1(0-1)(5)$

$= 0$

$w_2^{new} = w_2^{old} + \eta(y - \hat{y}) x_2$

$= 0.3 + 0.1(0-1)(2)$

$= 0.1$

$w_3^{new} = w_3^{old} + \eta(y - \hat{y}) x_3$

$= 0 + 0.1(0-1)(1)$

$= -0.1$

row 4 (10, 5, 1) ; $\eta = 0.1$

$w_1 = 0 \quad w_2 = 0.1 \quad w_3 = -0.1 \quad y = 1$

$z_4 = w_1 x_1 + w_2 x_2 + w_3 x_3$

$= 10(0) + 0.1(5) + (-0.1)(1)$

$= 0.4$

$z_4 > 0$

$\boxed{\hat{y} = 1}$

$y = \hat{y}$

row 5 (7, 3, 1) ; $\eta = 0.1$

$w_1 = 0 \quad w_2 = 0.1 \quad w_3 = -0.1 \quad y = 1$

$z_5 = w_1 x_1 + w_2 x_2 + w_3 x_3$

$= (0)(7) + (0.1)(3) + (-0.1)(1)$

$= 0.2$

$z_5 > 0$

$\boxed{\hat{y} = 1}$

$\boxed{y = \hat{y}}$

A movie st
wants to c
into categor
Romance w

Input Feat
Action Sc

Romance
romantic

Target V
genere (

Training
Movie ID

1
2
3
4
5
6
7
8